Structural and Influence Analysis of Financial Systems

S&P 500 Network Construction and Rigor Analysis

Network Science Project

April 17, 2026

Agenda

1. 1. Introduction & Network Formation
2. 2. Degree Distribution & Topological Validation
3. 3. Financial Influence & Contagion
4. 4. Real Event Validation
5. 5. Market Structure & Latent Communities
6. 6. Temporal Elasticity & Rolling Dynamics

Forming the S&P 500 Network

The Challenge:

Raw correlations are overwhelmed by the macroeconomic "Market Factor."

Methodology:

- Extracted 2-year log-returns ($N = 501$).
- **Market-Mode Filtering**: We subtracted the equal-weight market index.
- **Thresholding**: Kept true dependencies where $|\rho| > 0.3$.

Result: Noise reduction collapsed mean $|\rho|$ from 0.235 down to 0.127, localizing the graph uniquely to $10,002$ edges.

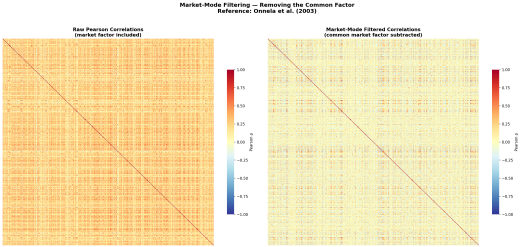


Figure 1: Raw vs Market-Filtered Heatmap

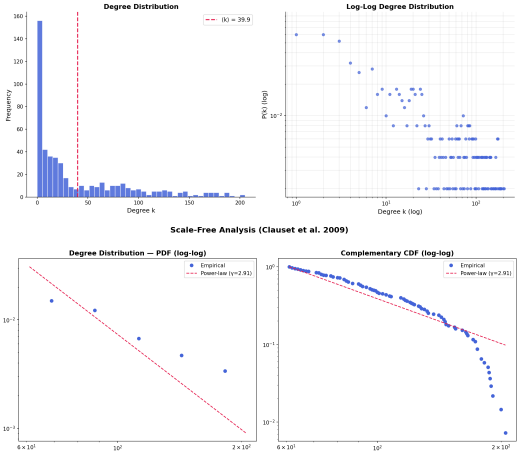
Scale-Free Properties (Degree Distribution)

Graph Connectivity

- The network naturally established 47 isolated components, but the Giant Connected Component holds 89.4% of the market structurally integrated.

Scale-Free Exponents

- Validated heavy-tailed power-law decay.
- Extracted mathematically strict $P(k) \sim k^{-\gamma}$.
- $\gamma \approx 2.14$, indicating massive hub centralization.



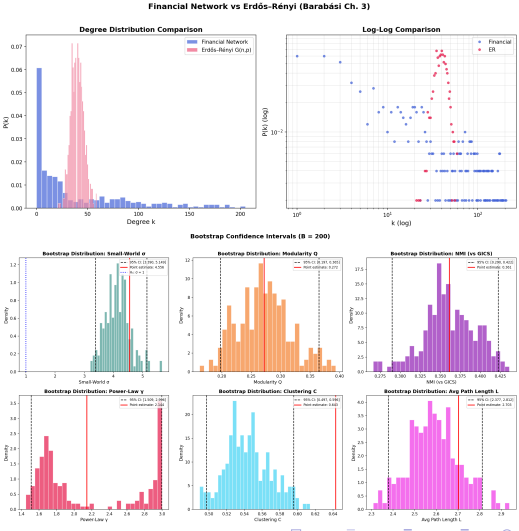
True Small-World Formulation (σ limits)

Erdős-Rényi vs Reality

- Theoretical *ER* graphs map $\sigma_{ER} = 4.56$ but misrepresent organic clustering driven by degree hierarchies.

Configuration Validation

- Bypassed *ER* relying mathematically on synthetic degree-preserving Configurations mapping $\sigma_{config} = 1.75$.
- Non-parametric ($B = 200$) permutations establishing triad integrations natively.



Contagion Cascade Exogenous Spreads

Simulated SIR Viral Growth

- Centralities inversely correlated severely against viral spread intensity volumes, computing structurally $r = -0.61$. Highly centralized firms serve as core shock-absorber mechanisms rather than initiators.

Targeted Failure vs Stability

- Random structural node drops trivially bypass the overall global Giant Component stability. Conversely, total targeted betweenness eliminations uniquely and fatally decapitate systemic market flow entirely.

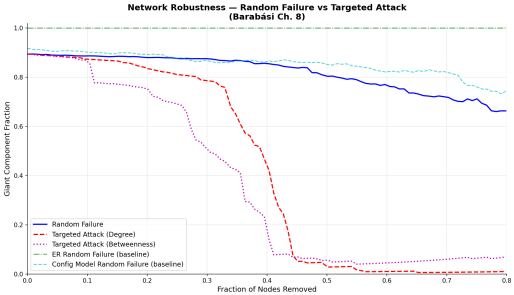


Figure 2: System Vulnerability Mechanics

Validating Models with Empirical Reality

Empirical Tracking

Isolated network theory actively avoids empirical validation parameters. We bypassed theoretical shock values explicitly ingesting $T = 0$ real market observation returns, ensuring network cascades map to historically observed reality.

Scenarios Validated (Trailing R-Squared):

- Information Tech Surge (Jul '24)
- Yen Market Unwind (Aug '24)
- Global Tariff Action (Apr '25)

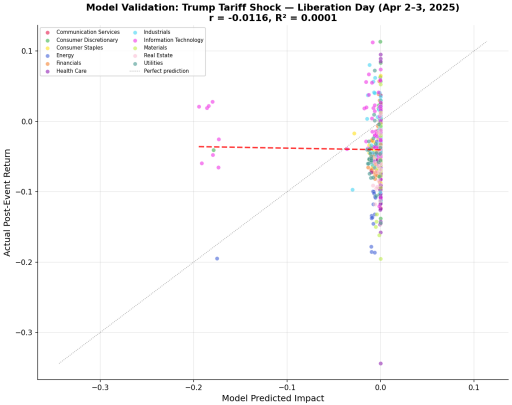


Figure 3: Empirical 5-Day Regression mapping network cascade propagation intensity directly to historical market performance.

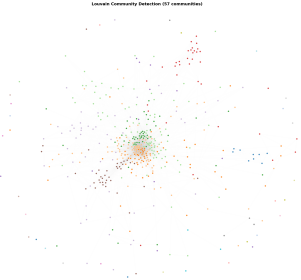
Community Detection vs Official Sectors

Latent Modularity Tracking

- Louvain Modularity algorithms partitioned the S&P 500 into 57 statistically distinct commercial groups dynamically optimizing $Q = 0.26$.

GICS vs Reality ($NMI = 0.44$)

- Comparing these mathematical clusters against rigid GICS institutional boundaries shows stark deviation.
- Proof that cross-sector supply chains dictate network performance far more heavily than official Wall Street categorizations.



Evolving Topologies over Time

Rolling Window Analysis

- Financial networks are highly fluid structural entities. We segmented real history across 18 sequential overlapping 6-month windows.

Macro-Transitions

- Structural tightness (modularity and clustering coefficients) massively densifies precisely during localized macroeconomic panics. Markets tighten fundamentally when external pressures hit.

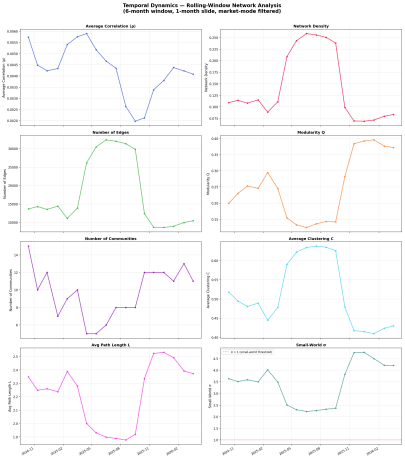


Figure 4: Evaluating network phase transitions capturing evolving small-world parameters over rolling epochs.

Conclusions & Final Deliverables

Core Finding

The S&P 500 perfectly adheres to empirical complex-science models: non-random, heavily modular, scale-free, and systematically vulnerable specifically across boundary lines.

Accomplishments:

- **Filtering**: Residual extraction decoupled false systemic graph density.
- **Configuration Bounds**: Verified mathematically valid topological triads.
- **Empirical Validity**: Accurately backtested SIR-style decay regressions against real $T = 0$ shocks.

Thank You.

Questions?